

Problem Statement or Requirement:

A client’s requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

- 1.) Identify your problem statement
- Stage1: Domain – Machine learning
- Stage2. Supervised Learning
- Stage3: Regression
- 2.) Tell basic info about the dataset (Total number of rows, columns)
- 6 columns and 1338 rows. Input dataset =age, sex, BMI, children and smoker. Output dataset=charges
- 3.) Mention the pre-processing method if you’re doing any (like converting string to number – nominal data)
- dataset=pd.get_dummies(dataset, dtype=int ,drop_first=True)
- This coding I have used to convert categorical datas into nominal datas.
- 4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

SVM

S.No	Hyper parameter	poly (r value)
1	C=1000	0.856648

Random Forest

S.No	Criterain	Estimator	R_score	After standardisation
1	friedman_mse	100	0.854051	0.855143

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Multiple linear regression

R value **0.789479**

Destination Tree

S.No	Criterain	Splitter	R_score
1	friedman_mse	random	0.751971
2	friedman_mse	best	0.704554
3	squared_error	random	0.709793
4	squared_error	best	0.68339
5	absolute_error	random	0.708826
6	absolute_error	best	0.68632
7	poisson	random	0.701266
8	poisson	best	0.71022

SVM

S.No	Hyper parameter	Linear(r value)	rbf (r value)	poly (r value)	Sigmoid (r value)
1	C=1000	0.764931	0.810206	0.856648	0.28747
2	C=100	0.6288792	0.320031	0.617956	0.52761

Random Forest

S.No	Criterain	Estimator	R_score	After standardisation
1	friedman_mse	100	0.854051	0.855143
2	friedman_mse	50	0.850071	0.851119
3	squared_error	100	0.85383	0.854954
4	squared_error	50	0.848932	0.850911
5	absolute_error	100	0.852009	0.853186
6	absolute_error	50	0.852665	0.854123
7	poisson	100	0.852633	0.853588
8	poisson	50	0.849107	0.850325

5.) Mention your final model, justify why u have chosen the same.

Random forest creates model with r_value 0.855143. As r_value (evaluation metrics)is higher than other models, I took this as a good model and send it to deployment phase.